

ANALYSTLAB AFRICA
MACHINE LEARNING INTERNSHIP PROGRAMME

WEEK 3 BUSINESS REPORT
CUSTOMER CHURN PREDICTION

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1. Executive Summary

This report presents my Week 3 machine learning project on predicting customer churn using the Telco Customer Churn dataset. The main objective of my work was to develop classification models that can identify customers who are likely to leave a telecommunications service and to determine which model is most suitable for the business problem.

I continued from the data understanding and preparation completed in the previous weeks. The prepared dataset contained 7,043 customer records and 31 predictor variables, with Churn used as the target variable. I divided the data into training and testing sets using an 80/20 stratified split, producing 5,634 training observations and 1,409 testing observations.

I developed four classification models: Logistic Regression, Decision Tree, Random Forest and XGBoost. The models were evaluated using Accuracy, Precision, Recall, F1 Score and ROC-AUC, together with confusion matrices.

Random Forest produced the highest accuracy at 78.21% and the highest precision at 61.84%. However, XGBoost produced the highest recall at 81.02%, the highest F1 Score at 62.54%, and the highest ROC-AUC at 84.30%. Because the purpose of churn prediction is to identify customers who are likely to leave so that action can be taken before they leave, I recommend XGBoost as the preferred model, subject to further validation and optimization.

2. Modelling Approach

2.1 Problem Definition

I formulated the customer churn problem as a binary classification problem. The target variable, Churn, indicates whether a customer leaves the service. A value of 0 represents a customer who does not churn, while a value of 1 represents a customer who churns.

The purpose of the model is to learn patterns from existing customer information and use those patterns to estimate whether another customer is likely to churn.

2.2 Data Preparation

I used the processed dataset from my Week 2 work as the foundation for modelling. Categorical variables had been converted into numerical variables through one-hot encoding, while relevant numerical variables were retained. The predictors included tenure, MonthlyCharges, TotalCharges, TotalServices, SeniorCitizen, contract categories, internet-service categories, payment-method categories and customer-service variables. The Churn column was separated from the predictor variables before model training.

2.3 Train-Test Split

I used an 80/20 train-test split with stratification. The training set contained 5,634 observations, while the testing set contained 1,409 observations.

I selected the 80/20 split because it provides the models with enough observations for learning while keeping an independent test set for evaluating performance on unseen customers. I also used stratification because the churn classes were imbalanced. This kept the proportion of churn and non-churn customers approximately the same in both the training and testing datasets.

3. Algorithms Selected

3.1 Logistic Regression

I selected Logistic Regression as a baseline classification model. It is relatively simple, computationally efficient and easier to interpret than many ensemble models. It provides a useful benchmark against which the more complex models can be compared.

3.2 Decision Tree

I selected a Decision Tree because it can model nonlinear relationships and produces decision rules that are relatively easy to understand. Its main limitation is that a single tree can be unstable and may not generalize as well as ensemble methods.

3.3 Random Forest

I selected Random Forest because it combines multiple decision trees to produce a more robust prediction. I used 100 trees and balanced class weights to account for the imbalance between churn and non-churn customers. Random Forest is useful for structured customer data because it can capture nonlinear relationships and interactions between variables.

3.4 XGBoost

I selected XGBoost because gradient boosting is effective for structured tabular datasets and can capture complex relationships between predictors and customer churn. I also accounted for class imbalance using a scale-positive-weight approach.

4. Evaluation Results

The results from my four models are summarized below.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	73.81%	50.43%	78.34%	61.36%	84.17%
Decision Tree	73.60%	50.26%	50.80%	50.53%	66.37%
Random Forest	78.21%	61.84%	46.79%	53.27%	82.22%
XGBoost	74.24%	50.92%	81.02%	62.54%	84.30%

4.1 Accuracy

Random Forest achieved the highest accuracy at 78.21%. This means it correctly classified the largest overall proportion of customers in the test set.

4.2 Precision

Random Forest also achieved the highest precision at 61.84%. This means that, among the customers it predicted as likely to churn, it had the highest proportion of actual churners.

4.3 Recall

XGBoost achieved the highest recall at 81.02%, followed by Logistic Regression at 78.34%. Recall is particularly important because a customer who is actually going to churn but is not identified by the model may be lost before the company has an opportunity to intervene.

4.4 F1 Score

XGBoost achieved the highest F1 Score at 62.54%. The F1 Score provides a balance between precision and recall and is useful when both types of classification error matter.

4.5 ROC-AUC

XGBoost achieved the highest ROC-AUC at 84.30%, narrowly outperforming Logistic Regression at 84.17%. This indicates that XGBoost had the strongest overall ability among the tested models to distinguish between churn and non-churn customers across different classification thresholds.

5. Model Comparison

The four models produced different strengths and weaknesses. Random Forest was the strongest model when accuracy and precision were considered, while XGBoost was strongest when recall, F1 Score and ROC-AUC were considered.

Model	Main Strength	Main Limitation	Overall Assessment
Logistic Regression	Strong recall and interpretability	Less effective for complex nonlinear patterns	Good baseline
Decision Tree	Easy to interpret	Weakest predictive performance	Not preferred
Random Forest	Highest accuracy and precision	Low churn recall	Strong alternative
XGBoost	Highest recall, F1 and ROC-AUC	More complex to interpret	Recommended

I recommend XGBoost as the preferred model for this particular business problem. Although Random Forest has higher accuracy, XGBoost identifies a much larger proportion of the customers who actually churn. For a retention-focused business, detecting potential churners is more important than simply maximizing overall classification accuracy.

6. Feature Importance

The feature-importance analysis helped me identify the customer characteristics that contributed most to the predictive performance of the tree-based models.

The most important predictors included Contract_Two year, Contract_One year, InternetService_Fiber optic, InternetService_No, PaymentMethod_Electronic check, StreamingMovies_Yes, tenure, MonthlyCharges and TotalCharges.

Contract-related variables were particularly important. This suggests that the type and duration of a customer's contract contain useful information for distinguishing customers who are likely to churn

from those who are likely to remain.

Tenure was also important. This indicates that the length of a customer's relationship with the company provides useful predictive information. Newer customers may require additional attention because their relationship with the company is less established.

Internet service and payment method also contributed strongly to prediction. These variables should therefore be considered when designing customer-retention strategies.

Feature importance describes predictive contribution rather than causation. A feature being important to the model does not prove that changing that feature will directly cause churn to increase or decrease.

7. Business Recommendations

7.1 Develop a Churn-Risk Identification System

I recommend using the selected model to assign each customer a churn probability. Customers can then be grouped into low-, medium- and high-risk categories so that retention resources can be focused on customers who need them most.

7.2 Prioritize Early Identification of Potential Churners

Since XGBoost achieved an 81.02% recall, it provides a useful basis for identifying a large proportion of customers who are likely to churn. The business should use these predictions to contact high-risk customers before they leave.

7.3 Focus on Contract-Based Retention

Because contract variables were among the most important predictors, the company should examine churn behaviour under different contract types. Where appropriate, it could offer loyalty benefits, contract-upgrade incentives or bundled services to encourage longer customer relationships.

7.4 Improve Early-Customer Engagement

Tenure was an important predictor, so customer engagement should be strengthened during the early stages of the customer relationship. Onboarding support, satisfaction checks and proactive customer service could help identify problems before they lead to churn.

7.5 Investigate High-Risk Service Segments

The importance of internet-service variables suggests that the company should investigate whether differences in service quality, pricing, customer expectations or technical support are associated with higher churn.

7.6 Use Targeted Retention Campaigns

Rather than applying the same retention strategy to every customer, the company should combine churn probability with customer characteristics to create targeted interventions. This can help reduce unnecessary discounts and make retention spending more efficient.

8. Next Steps

8.1 Hyperparameter Tuning

The next stage should involve tuning the model parameters, especially for XGBoost and Random Forest. Grid Search, Randomized Search or other optimization techniques can be used to determine whether performance can be improved.

8.2 Cross-Validation

Cross-validation should be performed to determine whether the observed results are consistent across different subsets of the training data and to obtain a more reliable estimate of model generalization.

8.3 Classification Threshold Optimization

The default classification threshold should not automatically be treated as optimal. Different thresholds should be evaluated based on the financial cost of missing a potential churner compared with contacting a customer who would have remained.

8.4 Model Explainability

Explainability techniques such as SHAP can be introduced to show why individual customers receive high churn-risk predictions. This would make the model easier for business stakeholders to understand and trust.

8.5 Deployment

After additional validation, the selected model can be integrated into a customer-management or business-intelligence system. New customer data can then be processed through the same preprocessing workflow and passed to the model to generate churn probabilities.

8.6 Continuous Monitoring

After deployment, model performance should be monitored regularly. Accuracy, recall, ROC-AUC, data drift and changes in customer behaviour should be tracked. The model should be retrained when its performance deteriorates or when the underlying customer population changes significantly.

9. Conclusion

In this Week 3 project, I developed four machine learning classification models to predict customer churn: Logistic Regression, Decision Tree, Random Forest and XGBoost.

The evaluation showed that Random Forest achieved the highest accuracy at 78.21% and the highest precision at 61.84%. However, XGBoost achieved the highest recall at 81.02%, F1 Score at 62.54%, and ROC-AUC at 84.30%.

Based on the business objective of identifying customers who are likely to leave, I recommend XGBoost as the preferred model for further development. Its high recall means that it can identify a larger proportion of actual churners, giving the business more opportunities to intervene before customers leave.

The feature-importance analysis also showed that contract type, tenure, internet service, payment method and customer charges are important predictive variables. These findings can support more targeted and data-driven customer-retention strategies.

Before production deployment, I would carry out further hyperparameter tuning, cross-validation, threshold optimization, explainability analysis and business-cost evaluation. This would help ensure that the final model is not only technically effective but also useful and sustainable in a real business environment.